

ENR 1.6 ATS SURVEILLANCE SERVICES AND PROCEDURES

1.6.1 Primary Radar

1. Primary radar equipment are located at:

Location	Coordinates	Range
Clark	151051.06N 1203351.92E	60 NM
Kalibo	114809.00N 1221305.00E	60 NM
Mactan	101832.00N 1235905.00E	60 NM
Manila *	143027.94N 1210125.77E	60 NM
Manila 2	143015.00N 1210110.00E	60 NM

* Note: From Radial 355 clockwise to Radial 009 in the vicinity of CABANATUAN and beyond, unusable at 10000 FT and below.

2. Application of Radar Control Service

2.1 Radar identification is achieved according to the provisions specified by ICAO.

2.2 Radar control service is provided in controlled airspace to aircraft operating within the Manila TMA, Clark TMA, Mactan TMA, Bacolod TMA, Davao TMA and the Kalibo TMA and certain areas under the jurisdiction of the Manila Area Control Center. This service may include:

- a. Radar separation of arriving, departing and enroute traffic;
- b. Radar monitoring of arriving, departing and enroute traffic to observe and note any significant deviation from the normal flight path, airway or route. This includes aircraft position relative to approach fixed and major obstructions;
- c. Radar vectoring to provide course guidance;
- d. Traffic information on other aircraft observed on the radar display, which may be in such proximity to its position or intended route of flight to constitute a hazard;
- e. Information on radar observed weather and chart areas;
- f. Assistance to aircraft in distress and VFR aircraft in weather difficulty;
- g. Information to assist in the navigation of aircraft; and
- h. Assistance to aircraft crossing controlled airspace.

2.3 Separation Standard

2.3.1 The horizontal separation minimum based on radar shall be five (5) NM.

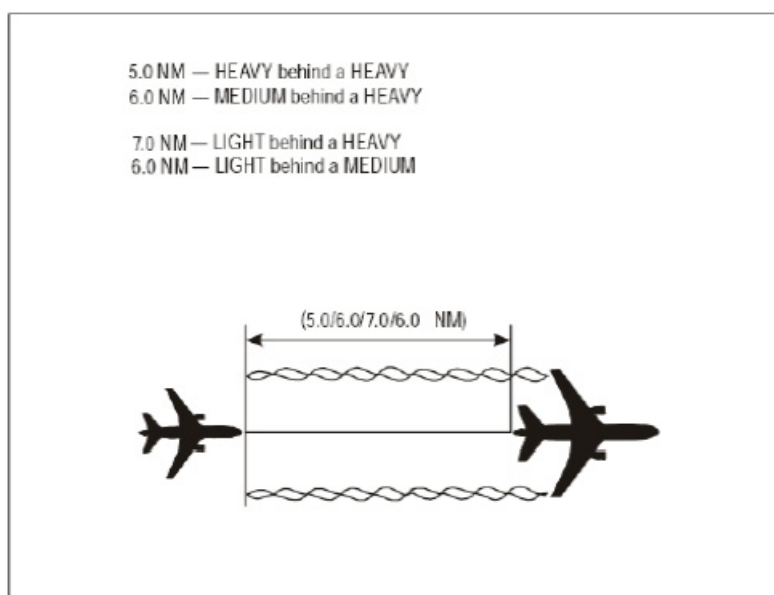
2.3.2 The separation minimum based on radar to be applied shall be according to the capability of the particular radar system or sensor to accurately identify the aircraft position in relation to the center of a position symbol, Primary Surveillance Radar (PSR) blip, Secondary Surveillance Radar (SSR) response and taking into account factors which may affect the accuracy of the radar-derived information, such as aircraft range from the radar site and the range scale of the situation display in use.

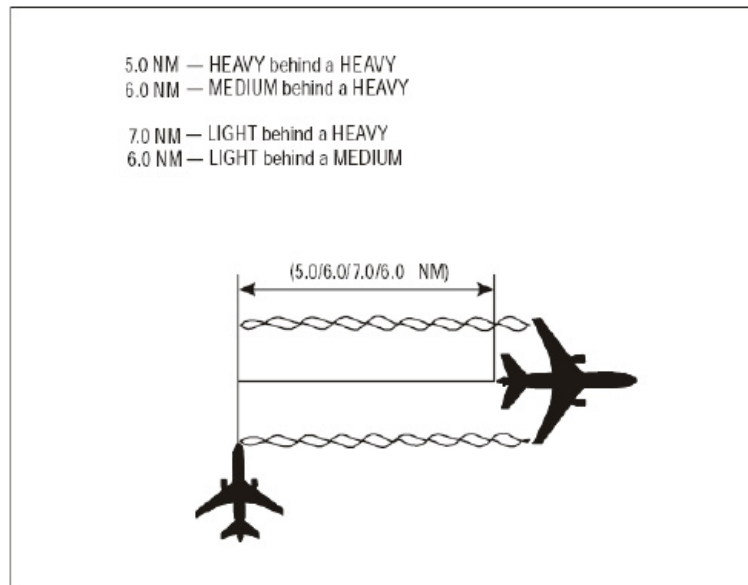
- 2.3.3 The following distance-based wake turbulence separation minima shall be applied to aircraft being provided with an ATS surveillance service in the approach and departure phases of flight in the circumstances given in 2.3.4.

Aircraft Category		
Preceding Aircraft	Succeeding Aircraft	Distance-based wake turbulence separation minima
HEAVY	HEAVY	5.0 NM
	MEDIUM	6.0 NM
	LIGHT	7.0 NM
MEDIUM	LIGHT	6.0 NM

- 2.3.4 The minima set out in 2.3.3 shall be applied when:

- an aircraft is operating behind another aircraft at the same altitude or less than 1000 FT below; or
- both aircraft are using the same runway, or parallel runways separated by less than 2500 FT; or
- an aircraft is crossing behind another aircraft, at the same altitude or less than 1000 FT below.





2.4 Identification Procedures

2.4.1 Radar will establish positive identification of aircraft before taking any control action based on radar display. Pilot will be informed when his aircraft has been radar identified. Identification will be established by using any of the following methods:

- Observing a target of departing aircraft within one (1) NM of the take-off runway;
- Requiring an aircraft to make one or more turns of thirty (30) degrees or more and observing that the turns made are consistent with the given instructions;
- Observing a target whose position corresponds to a fix or visual reporting point as reported by an aircraft and the observed track is consistent with the reported heading or route of flight; and
- Observing identification display of aircraft equipped with transponder when activated to the required mode and code.

2.5 Landing Priority

2.5.1 Radar controlled traffic executing a specific type of approach has no landing priority over other traffic during VMC. The control tower handles all aircraft, regardless of the type of flight plan, on a “first come, first served” basis. In the interest of safety, it may be necessary for the air traffic controller to provide different landing sequence to enable the pilot to adjust his flight path.

2.6 Termination of Radar Service

2.6.1 Aircraft being provided with radar service shall be informed immediately when, for any reason, radar service cannot be continuously provided. In such event, the radar approach controller shall issue instructions to revert to non-radar standard separation.

2.6.2 When transfer of radar controlled aircraft is to be made, coordination with non-radar facility shall be effected. Non-radar separation shall be established by a radar controller before an aircraft under radar control reaches the limits of non-radar area of responsibility.

3. Radar and Air-Ground Communications Failure Procedures

3.1 Radar Failure

- 3.1.1 In the event of radar failure or loss of radar identification, instructions will be issued as soon as practicable for the provision of non-radar separation and for the pilot to communicate with the appropriate ATS unit.

3.2 Radio Failure

- 3.2.1 In the event of radio failure, radar controller will establish whether the aircraft radio receiver is working by instructing the pilot to carry out a turn or turns. If the turns are observed, the radar controller will continue to provide radar service to aircraft.
- 3.2.2 If the aircraft's radio is completely unserviceable, the pilot should carry out the procedures for radio failure in accordance with ICAO provisions. If radar identification has already been established, the radar controller should vector other aircraft clear of its track until such time the aircraft leaves radar cover.
- 3.2.3 The following lost communications procedure phraseology is standard in radar ATC clearance and shall not be given by the air traffic controller unless otherwise requested by the pilot concerned:

"IF NO TRANSMISSIONS ARE RECEIVED FOR ONE MINUTE MAKE A VISUAL APPROACH, IF UNABLE PROCEED TO THE VOR, MAINTAIN 4,000 FEET OR LAST ASSIGNED ALTITUDE WHICHEVER IS HIGHER AND CONTACT TOWER IMMEDIATELY."

3.3 Aircraft Radio Transmitter Failure

- 3.3.1 If two-way communication is lost with an aircraft, the air traffic controller should determine whether or not the aircraft's receiver is functioning by instructing the aircraft on the channel so far used to acknowledge by making a specified maneuver and by observing the aircraft's track, or by instructing the aircraft to operate IDENT or to make SSR code transmission changes.

Note: Transponder-equipped aircraft experiencing radio communication failure will operate the transponder on Mode A Code 7600.

- 3.3.2 If the action prescribed in 3.3.1 is unsuccessful, it shall be repeated on any other available channel on which it is believed that the aircraft might be listening.
- 3.3.3 In both cases covered by 3.3.1 and 3.3.2 any maneuvering instructions shall be such that the aircraft would regain its current cleared track after having complied with the instructions received.
- 3.3.4 Where it has been established by the action in 3.3.3 that the aircraft's radio receiver is functioning, continued control can be effected using SSR code transmission changes or IDENT transmissions to obtain acknowledgement issued to the aircraft.

3.4 Complete Aircraft Communication Failure

- 3.4.1 When a controlled aircraft experiencing complete communication failure is operating or expected to operate in an area and at flight levels where an ATS surveillance service is applied, separation specified in 3.3.1 may continue to be used.

Note 1: If the aircraft experiencing the communication failure is not identified, procedural separation shall be applied between identified aircraft and all unidentified aircraft observed along the expected route of the aircraft with the communication failure, until such time as it is known, or can safely be assumed, that the aircraft with radio communication failure has pass through the airspace concerned, has landed, or has proceeded elsewhere.

Note 2: "Unidentified aircraft" in the above paragraph refers to aircraft with SSR return.

- 3.5 Aircraft transponder failure in areas where the carriage of a functioning transponder is mandatory.
- 3.5.1 When an aircraft experiencing transponder failure after departure is operating or expected to operate in an area where the carriage of a functioning transponder with specified capabilities is mandatory, the ATC units concerned should endeavour to provide for continuation of the flight to the aerodrome of first intended landing in accordance with the flight plan. However, in certain traffic situations, either in terminal areas or en-route, continuation of the flight may not be possible, particularly when failure is detected shortly after take-off. The aircraft may then be required to return to the departure aerodrome or to land at the nearest suitable aerodrome acceptable to the operator concerned and to ATC.

4. Voice and CPDLC Position Reporting Requirements (To be developed)

5. Graphic Portrayal of Area of Radar Coverage (To be developed)

1.6.2 Secondary Surveillance Radar (SSR)

Secondary Radar equipment are located at:

Location	Coordinates	Range
Aparri (Enroute SSR)	181810.00N 1213939.00E	250 NM
Bacolod	104622.00N 1230042.00E	230 NM
Clark	151051.06N 1203351.92E	110 NM (adjustable up to 220 NM)
Davao	070813.00N 1253859.00E	230 NM
Kalibo	114809.00N 1221305.00E	230 NM
Laoag (Enroute SSR)	181058.26N 1203206.76E	250 NM
Mactan	101832.00N 1235905.00E	230 NM
Manila *	143027.94N 1210125.77E	90 NM
Manila 2	143015.00N 1210110.00E	180 NM
Mt. Majic (Enroute SSR)	101932.00N 1234532.00E	250 NM
Palawan (Enroute SSR)	091403.00N 1180024.00E	250 NM
Tagaytay (Enroute SSR)	140523.78N 1205428.92E	250 NM
Zamboanga (Enroute SSR)	065521.00N 1220153.00E	250 NM

* Note: From Radial 355 clockwise to Radial 009 in the vicinity of CABANATUAN and beyond, unusable at 10000 FT and below.

1. Emergency Procedures

- 1.1 Except when encountering a state of emergency, pilots of aircraft operating within the Manila FIR shall transpond the SSR code last assigned to them by an ATS unit or, if no code has been previously assigned, to transpond on A0600 and Mode C. Pilots who have received specific code assignments shall maintain that setting unless otherwise instructed.

- 1.2 Pilots of aircraft in a state of emergency shall set the transponder to Mode A Code 7700.

2. **ATS Surveillance System Failure**

- 2.1 In the event of complete failure of the ATS surveillance radar system where air-ground communications remain, the controller shall plot the positions of all aircraft already identified, take the necessary action to establish procedural separation between the aircraft and, if necessary, limit the number of aircraft permitted to enter the area.

3. **Air-Ground Communication Failure and Unlawful Interference Procedures**

- 3.1 In the event of radio communications failure, the pilot of an aircraft shall set the transponder to Mode A Code 7600 and follow established procedures. Subsequent control of the aircraft shall be based on those procedures.
- 3.2 Pilots of aircraft subjected to unlawful interference shall endeavor to set the transponder to Mode A Code 7500, follow established procedures and notify the appropriate ATS unit of any significant circumstances associated therewith and any deviation from the current flight plan.

4. **SSR Code Assignment**

- 4.1 The following SSR codes shall be used within the Manila FIR:

International departures originating from the Manila FIR	2700-2777 6500-6577
International arrivals and overflights not conforming with Originating Regional Code Assignment Method (ORCAM)	0600-0677
Domestic Flights	4000-4077 4200-4277
Local Flights within Manila TMA	1000-1077
Local Flights within Mactan TMA	0300-0377
Local Flights within Clark TMA	0500-0577

5. **Voice and CPDLC Position Reporting Requirements (To be developed)**

6. **Graphic Portrayal of Area of SSR Coverage (To be developed)**

1.6.3 **Automatic Dependent Surveillance - Broadcast (ADS-B) (To be developed)**

1.6.4 **Other Relevant Information and Procedures**

1. **General**

- 1.1 Surveillance units are operated as an integral part of the Philippine ATS system and provide surveillance service to aircraft, to the maximum extent practicable, to meet international requirements. Many factors such as surveillance coverage, controller workload and equipment capabilities may affect these services, and the surveillance controller shall determine the practicability of providing, or continuing to provide, surveillance services in any specific case.
- 1.2 Pilots shall be advised on commencement and termination of radar services.
- 1.3 Three Scope (Three Sector) Operation of Manila Approach Control

1.3.1 Purpose

1.3.1.1 To decongest communications between pilot and controller in the Manila Approach frequencies and to define the responsibilities, procedures as well as coordination among the three (3) sectors Final/Departure Control (FMI) and Arrival Control/Approach Control consisting of the Manila South Sector (FMS) and Manila North Sector (FMN).

1.3.2 General

1.3.2.1 The Manila Approach Control provides surveillance approach control service within the Manila Terminal Control Area (TMA) boundaries. It coordinates with the various (adjacent) Air Traffic Control facilities, such as Manila ACC, Manila Tower, Clark Approach, and the Philippine Aeronautical Rescue Coordination Center (PARCC). It also receives and relays information related to IFR aircraft's flight status to airline companies and to Manila International Airport Authority (MIAA) as requested.

1.3.2.2 The area of responsibility and jurisdiction of the Manila Approach Control is the Manila TMA from 1500 FT up to FL150 per Letter of Agreement between the Manila Approach Control unit and the Manila ACC. Its responsibility includes the Manila Control Zones as defined by its lateral boundaries.

1.3.3 Specific

1.3.3.1 Operating Frequency

Working Positions	Primary Frequency (MHZ)	Secondary Frequency (MHZ)
FMI	124.4	127.7
FMS	124.8	127.7
FMN	119.9	127.7

1.3.3.2 Final/Departure Control (FMI)

1.3.3.2.1 The area of jurisdiction of the Final/Departure Control shall be the airspace defined with the coordinates, with vertical limits from 1500 FT to 7000 FT including the Manila Control Zone.

1.3.3.3 Arrival Control (FMS/FMN)

1.3.3.3.1 The area of jurisdiction of the arrival control shall be the airspace bounded by the lateral limits of the Manila TMA, including the 60 NM arc from Manila DVOR with vertical limits from 8000 FT up to and including FL150 except for the airspace and altitude relegated to the Final/Departure Control.

1.4 Manila Approach Sectorization

1.4.1 Three (3) Scope Operations

Name Lateral limits Vertical limits Class of Airspace	Unit providing service	Call sign Language Hours of operations	Frequency/Purpose
Functional Volume of Manila Inner (FMI) Sector 145421N 1212724E - 144012N 1213539E - 140601N 1203425E - 142025N 1202556E - 145421N 1212724E 7000 FT 1500 FT Class of Airspace: D	Manila APP	Manila DEP/Manila Final English 2300 - 1459	124.4 MHZ PRI FREQ 127.7 MHZ SRY FREQ

Name Lateral limits Vertical limits Class of Airspace	Unit providing service	Call sign Language Hours of operations	Frequency/Purpose
Functional Volume of Manila South (FMS) Sector 134524N 1202021E - 141315N 1203008E - 140601N 1203425E - 144012N 1213539E - 144723N 1213128E - 150031N 1215500E clockwise 60 NM radius including 144217.5N 1221947.9E (JOM DVOR/DME) - 134524N 1202021E <u>FL150</u> 8000 FT Class of Airspace: D	Manila APP	Manila APP/Manila ARR English 2300 - 1459	124.8 MHZ PRI FREQ 127.7 MHZ SRY FREQ
Functional Volume of Manila North (FMN) Sector 150031N 1215500E - 144723N 1213128E - 145421N 1212724E - 142025N 1202556E - 141315N 1203008E - 134524N 1202021E clockwise 60 NM radius excluding Clark TMA - 150031N 1215500E <u>FL150</u> 8000 FT Class of Airspace: D	Manila APP	Manila APP/Manila ARR English 2300 - 1459	119.9 MHZ PRI FREQ 127.7 MHZ SRY FREQ

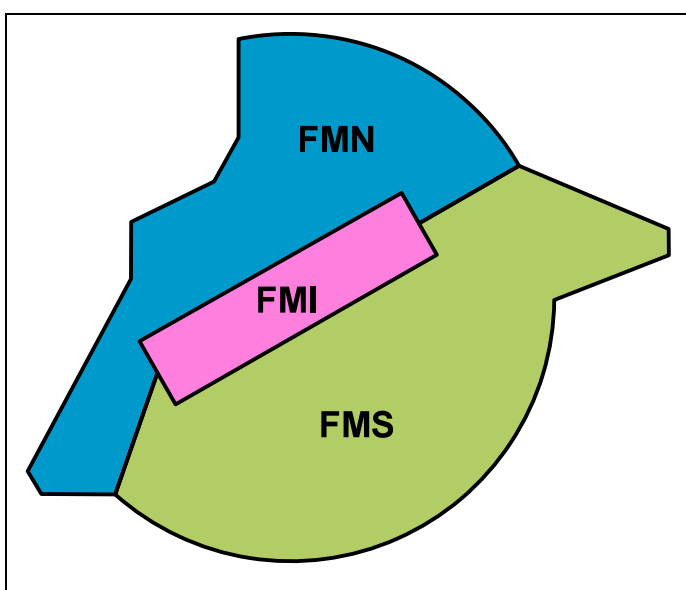


Figure 1: Three (3) Scope

1.4.2 Two (2) Scope Operations

Name Lateral limits Vertical limits Class of Airspace	Unit providing service	Call sign Language Hours of operations	Frequency/Purpose
Functional Volume of Manila Inner (FMI) Sector 145421N 1212724E - 144012N 1213539E - 140601N 1203425E - 142025N 1202556E - 145421N 1212724E <u>7000 FT</u> 1500 FT Class of Airspace: D	Manila APP	Manila DEP/Manila Final English 1500 - 1659 2000 - 2259	124.4 MHZ PRI FREQ 127.7 MHZ SRY FREQ
Functional Volume of Manila Outer (FMO) Sector (Merging of FMS and FMN) Functional Volume of Manila South (FMS) Sector 134524N 1202021E - 141315N 1203008E - 140601N 1203425E - 144012N 1213539E - 144723N 1213128E - 150031N 1215500E clockwise 60 NM radius including 144217.5N 1221947.9E (JOM DVOR/DME) - 134524N 1202021E Functional Volume of Manila North (FMN) Sector 150031N 1215500E - 144723N 1213128E - 145421N 1212724E - 142025N 1202556E - 141315N 1203008E - 134524N 1202021E clockwise 60 NM radius excluding Clark TMA - 150031N 1215500E <u>FL150</u> 8000 FT Class of Airspace: D	Manila APP	Manila APP/Manila ARR English 1500 - 1659 2000 - 2259	124.8 MHZ PRI FREQ 127.7 MHZ SRY FREQ

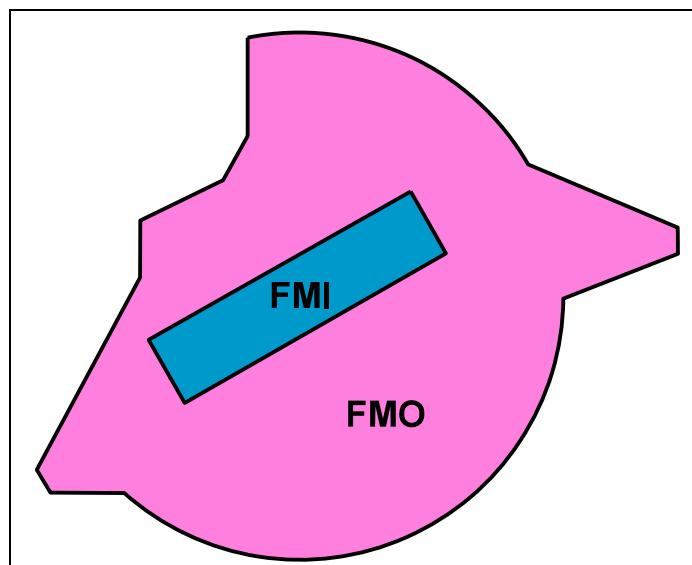


Figure 2: Two (2) Scope

1.4.3 One (1) Scope Operation

Name Lateral limits Vertical limits Class of Airspace	Unit providing service	Call sign Language Hours of operations	Frequency/Purpose
Manila TMA Boundary FM 152932N 1204903E then a clockwise arc radius 60 NM centered on 143028.7N 1210118.2E (MIA DVOR/DME) - 150030N 1215500E - 144555N 1223005E - 143955N 1223005E - 142949N 1220309E then a clockwise arc radius 60 NM centered on 143028.7N 1210118.2E (MIA DVOR/DME) - 134525N 1202021E - 134525N 1200305E - 135037N 1195953E - 143440N 1202353E - 144740N 1202353E - 145658N 1204320E - 150704N 1204903E - 152932N 1204903E FL150 1500 FT Class of Airspace: D	Manila APP	Manila APP English 1700 - 1959	124.4 MHZ PRI FREQ 127.7 MHZ SRY FREQ

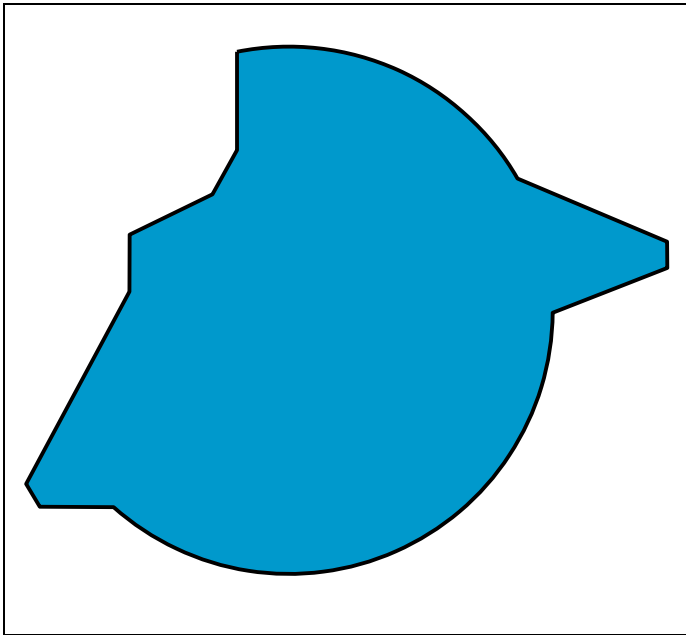


Figure 3: One (1) Scope

2. Remote Control Air-Ground (RCAG) Enroute VHF Communication facilities

Site Location	Coordinates	Frequencies (MHZ)
Aparri RCAG	181804.00N 1213941.00E	119.3 / 121.5
Cauayan Airport	165538.40E 1214509.55E	128.7
Davao RCAG	070813.00N 1253859.00E	132.2 / 121.5
Kalibo RCAG	114809.00N 1221305.00E	125.7
Laoag RCAG	181058.00N 1203209.00E	119.3 / 121.5
Manila Air Traffic Management Center (ATMC)	143047.00N 1210023.00E	120.5 / 125.7 / 132.7
Mt. Majic RCAG	101932.00N 1234532.00E	125.7 / 127.5 / 132.2 124.9 / 121.5
Palawan RCAG	091403.00N 1180024.00E	118.9 / 121.5
Puerto Princesa Airport	094425.39N 1184527.07E	118.9
Tagaytay Receiver Station	140540.00N 1205410.00E	132.7 / 128.7 / 121.5
Tagaytay Transmitter Station	140524.00N 1205429.00E	132.7 / 128.7 / 121.5
Zamboanga RCAG	065522.00N 1220155.00E	127.5 / 121.5

INTENTIONALLY
LEFT
BLANK